Project Report for Artificial Intelligence in Genomics (Spring 2026)

Comparing Interpretable Sequence Features and Protein Language Models for Missense Variant Classification

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Background and motivation

A *genetic variant* is a place in a person's DNA that differs from the human reference. We focus on *missense variants*: DNA changes that swap one amino acid for another in the protein; some are disease-causing (*pathogenic*) and others are tolerated (*benign*). **ClinVar** [2, 7] is our primary label source. **ESM-2** [3, 10] is a protein language model whose *residue embeddings* (1280-d for the 650M checkpoint) summarize evolutionary context, building on work showing masked language modeling captures mutational effect signal [4]. We ask, on a **gene-held-out** split, (i) how much signal lives in small interpretable feature sets and (ii) whether frozen embeddings improve cross-gene AUROC/AUPRC beyond those features: the tradeoff among interpretability, compute cost, performance, and generalization.

Methods

Data. We use NCBI's ClinVar `variant_summary.txt` (release dated **2026-05-04**, the snapshot pinned in `data/processed/clinvar_release.txt`) [6]. We restrict to GRCh38 germline single-nucleotide missense variants whose ClinVar review status maps to a clear binary label: pathogenic / likely-pathogenic vs. benign / likely-benign [5, 8]. We exclude conflicting and uncertain interpretations. Duplicates (same RefSeq transcript, position, ref AA, alt AA) are dropped, keeping the strongest review status. Table A2 reports the per-stage retention counts. We map each retained variant to a *reviewed* (Swiss-Prot) human protein sequence in UniProt UP000005640 [11], requiring that the residue at the variant position in the canonical sequence equals the HGVS-reported reference amino acid. After this verification we retain **193,483** variants across **14,893** genes (62,241 pathogenic / 131,242 benign). Figure A4 sketches the full pipeline.

Splits. To prevent *train–test contamination across variants of the same gene*, we use a **gene-disjoint** 70/10/20 train/val/test split (seed 42), stratified across genes by majority label so neither split is all-positive or all-negative. Final split sizes: **train 135,418 / val 19,539 / test 38,526** variants, with 10,425 / 1,489 / 2,979 unique genes respectively.

Models. We train **four** in-house predictors: two baselines on handcrafted features (logistic regression and random forest) and two neural heads on frozen ESM-2 residue embeddings (embedding-only and concatenated with handcrafted features), as summarized in Table 1. **AlphaMissense** [1] and **CADD** [9] are *external* reference scorers (not trained in this project; not targets to outperform); we evaluate them on the same gene-held-out test variants alongside our models (Phase 10; Table 2). The handcrafted vector per variant is exactly the five proposal families (20-d reference AA one-hot,

Table 1: **In-house models at a glance.** Neural heads keep ESM-2 650M frozen; training uses 135,418 gene-disjoint variants.

Model	Input	Architecture	Training
Logistic regression	62	L2 LR, balanced, C tuned on val AUPRC	Train split only
Random forest	62	500 trees, balanced, depth/leaf grid on val AUPRC	Train split only
ESM-2 head	1280	MLP $d \to 256 \to 1$, LayerNorm, dropout 0.2, GELU	AdamW + cosine; details in Appendix
Combined head	1342	Same MLP with $d = 1342$, concat(handcrafted, emb.)	Same as ESM-2 head

20-d alternate one-hot, BLOSUM62 score, 20-d local composition with window radius 7, normalized position; 62-d total). Tile handling for long proteins and optimization hyperparameters are deferred to the Appendix training-details section.

ESM-2 caveat (proposal-flagged). ESM-2 was pretrained on UR50, which contains many human proteins; this could inflate metrics on genes already well-represented in pretraining. We surface this caveat here and analyze it empirically in the appendix (Figure A7). **Computational simplicity** is reported per model in Appendix Table A3.

Evaluation. Metrics mirror Phase 9: AUROC, AUPRC, F1 at the val-tuned threshold, and 95% bootstrap CIs (1000 resamples). AlphaMissense and CADD are scored on the Phase 10 intersections with the test variants.

Results

Both empirical questions answered by the numbers in Table 2. The interpretable handcrafted random forest reaches test AUROC = 0.850 and AUPRC = 0.764 – roughly *nine-tenths of the ESM-2 head's AUROC* and *85% of its AUPRC* on a strict cross-gene test. A paired bootstrap on identical variants confirms the ESM-2 head's AUROC advantage over that forest ($\Delta = +0.089$, 95% CI $[0.085, 0.093]$, two-sided $p < 0.001$). Concatenating handcrafted features with embeddings yields a further increment (combined head: AUROC 0.954, +0.015 vs. embedding-only), indicating partial complementarity. AlphaMissense remains the strongest reference (AUROC 0.966 on $n = 37{,}470$ variants with scores); the ESM-2 head and CADD bracket it from below (Figure 1).

Gene-level and error structure. Per-gene AUROCs and confusion summaries (Figure 2) show the ESM-2 head's gene-level median above both baselines with a tighter spread, balanced precision/recall (0.800 / 0.841) versus LR (0.552 / 0.782) and RF (0.637 / 0.754). The combined head reaches the highest F1 among our models (0.850 at the val-tuned threshold).

Discussion

Direct numerical answers to the proposal's two questions. (i) *How much pathogenicity signal do simple interpretable features capture?* The L2 logistic regression reaches AUROC 0.800, and the random forest on the five headline families reaches AUROC 0.850 / AUPRC 0.764. A feature-family ablation at the Phase 7 RF hyperparameters (Appendix Figure A2) shows that

Table 2: **Test-set metrics with 95% bootstrap CIs.** Main test n = 38,526; AlphaMissense n = 37,470 (97.3% of test variants with scores); CADD n = 38,526 (Phase 10).

Model	**n**	**AUROC [95% CI]**	**AUPRC [95% CI]**	**F1 [95% CI]**	**Acc.**
Logistic regression	38,526	0.800 [0.796, 0.805]	0.668 [0.659, 0.677]	0.647 [0.641, 0.653]	0.708
Random forest	38,526	0.850 [0.846, 0.854]	0.764 [0.757, 0.771]	0.691 [0.684, 0.697]	0.768
ESM-2 head	38,526	0.939 [0.936, 0.941]	0.901 [0.897, 0.906]	0.820 [0.815, 0.825]	0.874
Combined head	38,526	**0.954 [0.952, 0.956]**	**0.924 [0.921, 0.928]**	**0.850 [0.845, 0.854]**	**0.896**
AlphaMissense (ref.)	37,470	0.966 [0.964, 0.967]	0.939 [0.936, 0.942]	0.872 [0.868, 0.876]	0.912
CADD (ref.)	38,526	0.928 [0.926, 0.931]	0.839 [0.832, 0.846]	0.808 [0.803, 0.813]	0.860

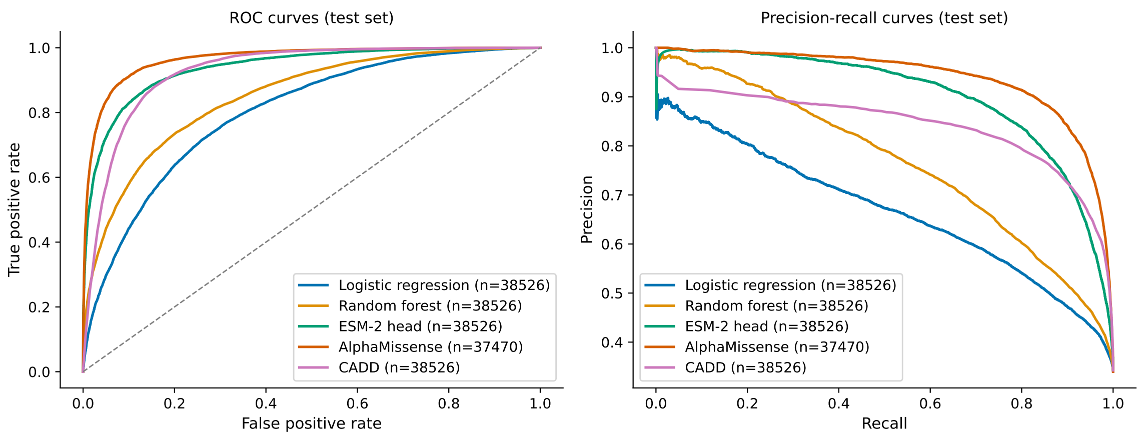

Figure 1: **Test-set ROC and PR curves.** All models on n = 38,526 gene-held-out test variants (AlphaMissense on 37,470 with score coverage). The ESM-2 head sits between the handcrafted baselines and AlphaMissense.

reference + alternate one-hot encodings plus the scalar BLOSUM62 term alone achieve test AUPRC 0.622, i.e. ∼**81%** of the full forest's AUPRC (0.764), with almost all remaining lift coming from the 20-d local composition window. (ii) *Does ESM-2 improve cross-gene generalization beyond handcrafted features?* Yes: the embedding-only head gains AUROC +0.089 and AUPRC +0.138 over the RF on mutually exclusive genes, and the paired bootstrap on identical variants confirms the AUROC contrast ($\Delta = +0.089$, 95% CI $[0.085, 0.093]$, two-sided $p < 0.001$). Concatenating embeddings with handcrafted vectors nudges performance upward again (Table 2), consistent with partially complementary information.

Where the forest and ESM-2 disagree. At validation-tuned thresholds, the random forest and ESM-2 head assign opposite hard labels on **9,552** test variants (**24.8%** of n = 38,526). Whenever binary predictions conflict, exactly one can match ClinVar; the ESM-2 head is correct on **6,801** of those disagreements versus **2,751** for the forest, qualitatively aligned with the AUROC gap, with scores and counts summarized in Appendix Figure A3 (and `rf_esm2_disagreement.csv`).

Tradeoffs, calibration, and limitations. Interpretability and disk footprint favor logistic regression; the forest remains expensive at inference; neural heads require a one-time embedding pass but rank best among our models. Reliability diagrams (Appendix Figure A1) show that all scorers are reasonably well centered, with the neural heads achieving the lowest Brier scores (Table A1). Beyond mapping and pretraining concerns already noted, **ClinVar** reflects database and ascertainment bias (over-studied genes, uneven depth [2]), pathogenic/likely-pathogenic vs.

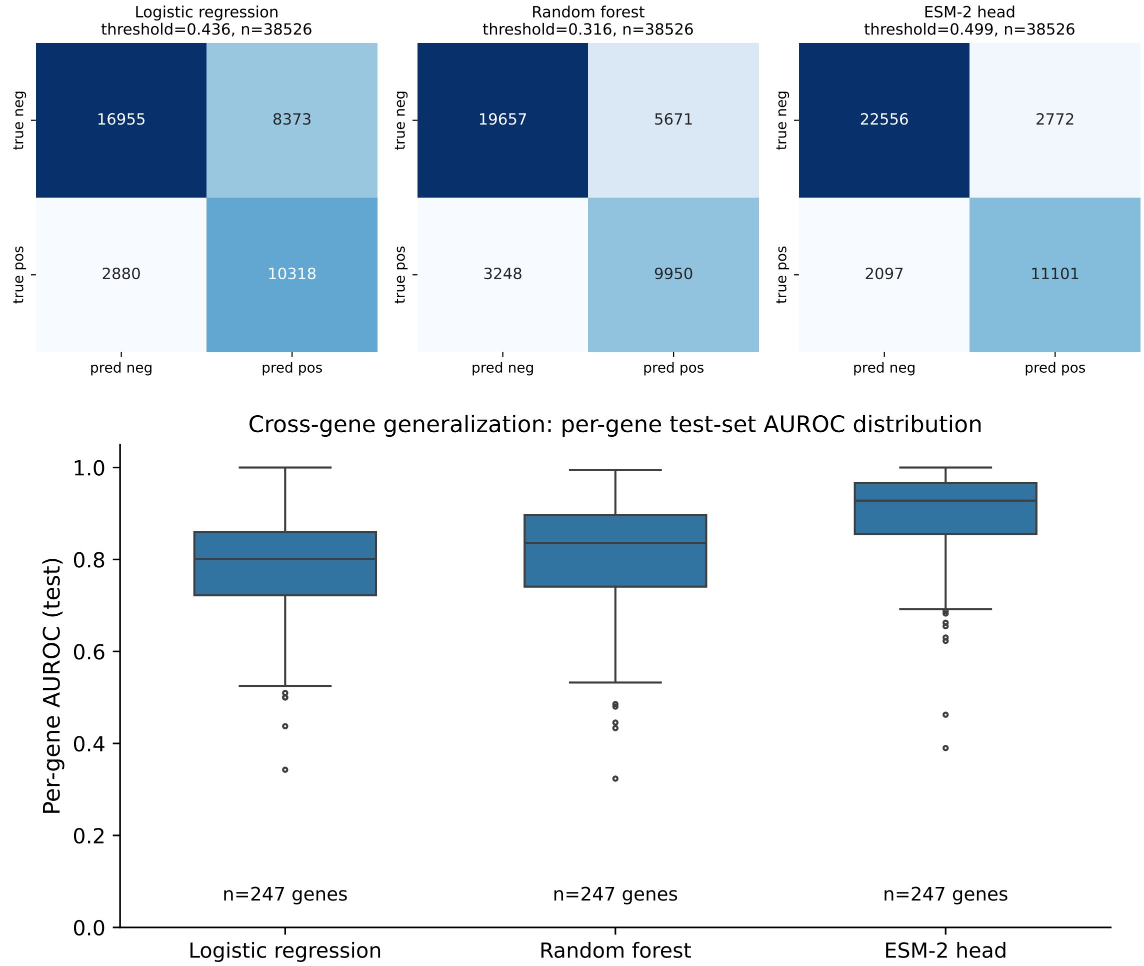

Figure 2: **Test-set confusion matrices (top) and per-gene AUROC distribution (bottom).** Top: val-tuned thresholds. Bottom: 247 genes with ≥ 5 pathogenic and ≥ 5 benign variants.

benign/likely-benign aggregation mixes label confidence, and gene-disjoint splitting limits blatant same-gene leakage but cannot remove shared biology or subtle annotation echoes across genes. External scores (AlphaMissense, CADD) are partly trained on overlapping literature labels and should be read as optimistic references. F1 uses val-tuned thresholds; the headline story is threshold-free AUROC/AUPRC.

Future work. End-to-end ESM-2 fine-tuning on ClinVar labels or structured clinical loss functions could probe how much of the current gap to AlphaMissense is fixable with data rather than architecture.

Supplementary materials

`supplementary_team19_gi2100_vsj7589.zip` bundles code, configs, pinned dependencies, `REPRODUCE.md`, data README, all tables/figures (including `pairwise_tests.csv`, `rf_esm2_disagreement.csv`, calibration, ablations), and the GitHub mirror `https://github.com/GeorgiosIoannouCoder/interpretable-missense-classification`. Run `make reproduce`; checkpoints stay resumable.

References

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[11] UniProt. Homo sapiens reference proteome (UP000005640). https://www.uniprot.org/proteomes/UP000005640, 2026. Official proteome page.

Appendix

Neural training details

Both the ESM-2 and combined heads are trained with AdamW (learning rate 10^{-3}, weight decay 0.01), batch size 512, BCE-with-logits with a positive-class weight $N_{\text{neg}}/N_{\text{pos}}$ estimated on the training split, and a cosine learning-rate schedule spanning all optimizer steps. Mixed precision (bf16) is enabled on CUDA when available via Hugging Face Accelerate (so the same code spans multi-GPU servers). Early stopping tracks validation AUPRC with patience five epochs; checkpoints are written every 500 steps and at epoch boundaries. For proteins exceeding ESM-2's 1022-token context we slide overlapping sequence windows, keep all variant-containing tiles, and select the tile whose center lies closest to the missense site before pooling the residue hidden state.

Table A1: **Calibration on the gene-held-out test set ($n = 38{,}526$).** Brier score and expected calibration error (ECE) with ten quantile bins; lower is better. Plots appear in Figure A1.

Model	Brier	ECE
Logistic regression	0.182	0.113
Random forest	0.148	0.040
ESM-2 head	0.093	0.049
Combined head	0.080	0.048

Table A2: **Filtering funnel.** Each row reports retained ClinVar variants after applying the labelled stage. The proposal's cited 2026-03-30 baseline (4.24 M variation records) is reproduced approximately at the GRCh38 stage with our 2026-05-04 snapshot.

Stage	Variants retained
Raw `variant_summary.txt` rows	8,978,110
After Assembly = GRCh38	4,455,427
After germline only	4,142,993
After single-nucleotide variant type	3,859,832
After missense parse	2,415,038
After P/LP or B/LB label only	210,038
After dedup by (transcript, AA pos, ref AA, alt AA)	208,490
After UniProt Swiss-Prot mapping with ref-AA verification	193,483

Table A3: **Computational-simplicity (efficiency) per model.** Inference latency measured per variant on the GH200 (CPU for sklearn models, GPU for the ESM-2 head – excluding the one-time 4 minutes of embedding extraction).

Model	Params	Disk size (MB)	Train time (s)	Inference (ms / variant)
Logistic regression	63	0.001	0.4	0.05
Random forest	500 trees	2,412.7	1.8	45.6
ESM-2 head	331,265	3.8	16.5	0.23
Combined head	347,261	4.0	11.4	0.21

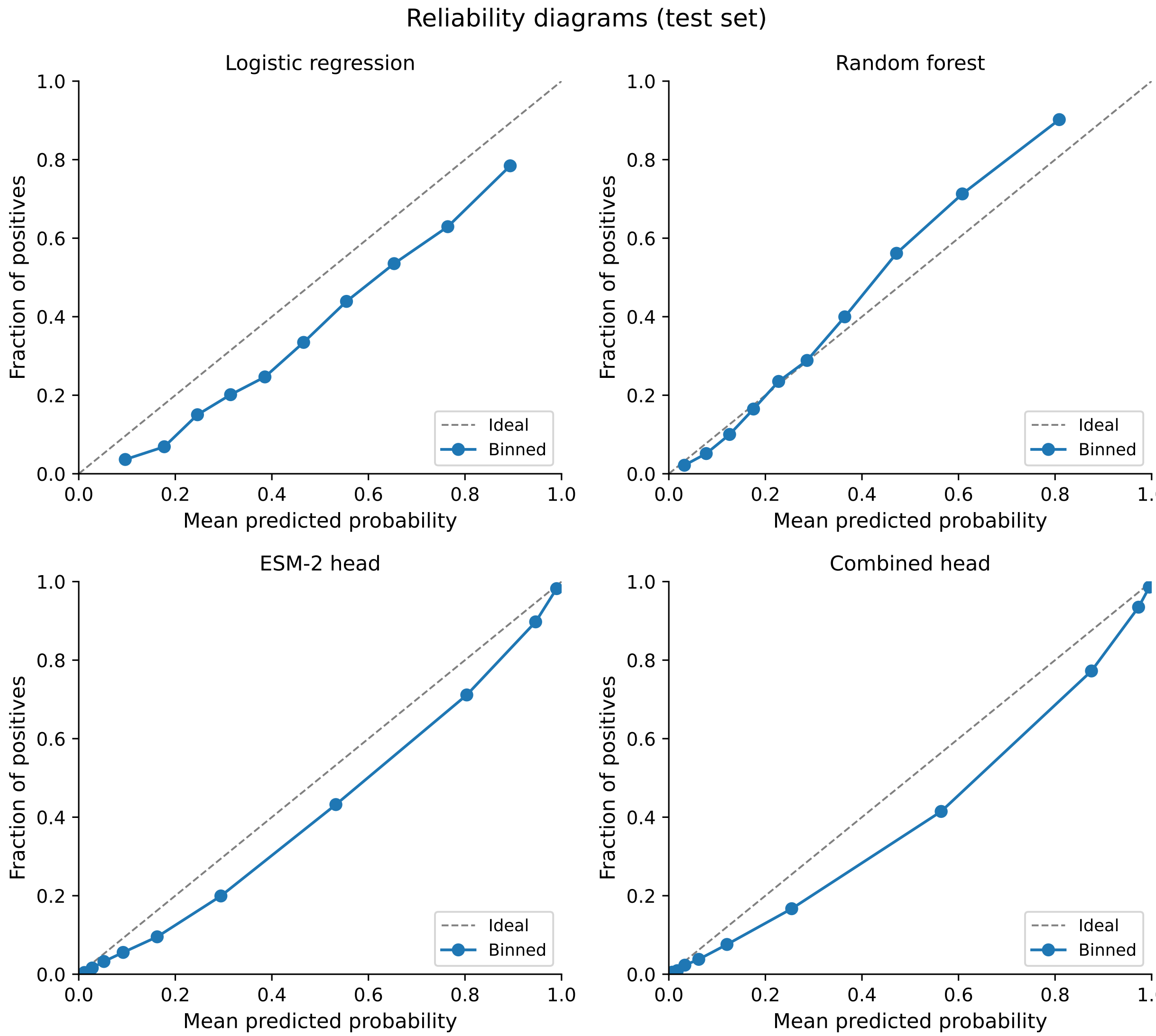

Figure A1: **Reliability diagrams.** Calibration curves for the four in-house scorers (10 quantile bins). Points should hug the diagonal under perfect calibration.

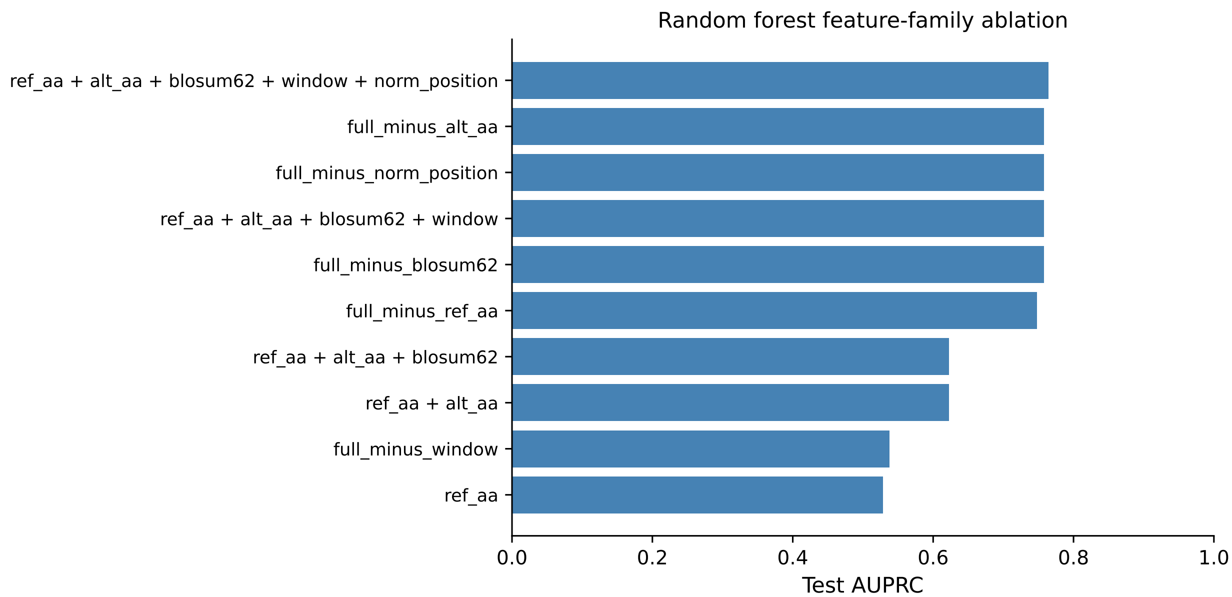

Figure A2: **Random forest feature-family ablation.** Test AUPRC when training only on cumulative feature subsets (top block) or removing one family from the full 62-d vector (bottom block) using the Phase 7 hyperparameters. The local window composition drives most of the lift beyond substitution-aware features.

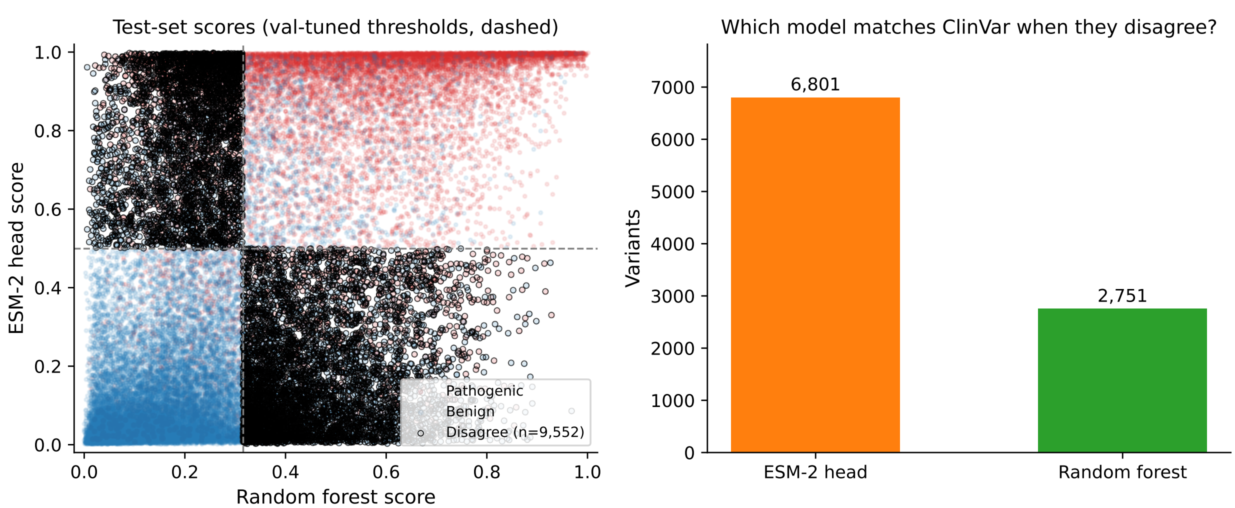

Figure A3: **Random forest vs. ESM-2 head disagreements.** *Left:* test-set scores with validation-tuned thresholds (dashed); black outlines mark variants where hard predictions differ. *Right:* among those disagreements ($n = 9{,}552$), how often each model matches the ClinVar binary label (6,801 vs. 2,751).

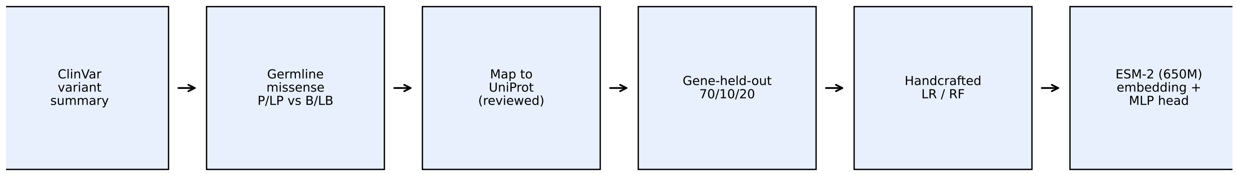

Figure A4: **Pipeline schematic.** Data flow from raw ClinVar to the gene-held-out evaluation.

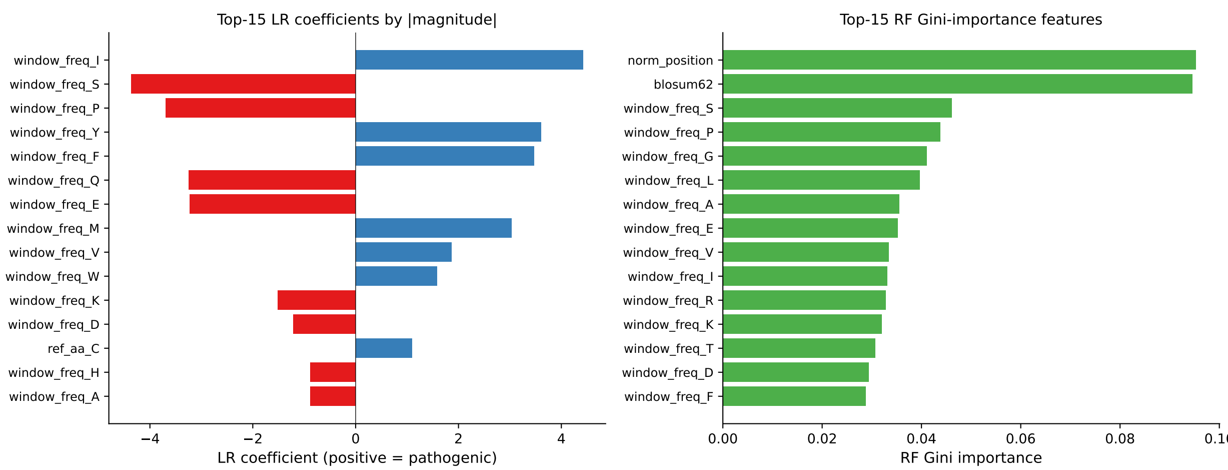

Figure A5: **Handcrafted-feature interpretability.** Top-15 LR coefficients (left) and top-15 RF Gini importances (right). The most influential features are alternate amino acids (Cys, Gly, Pro, Trp), the BLOSUM62 score, and one-hot reference amino acids – all biologically expected.

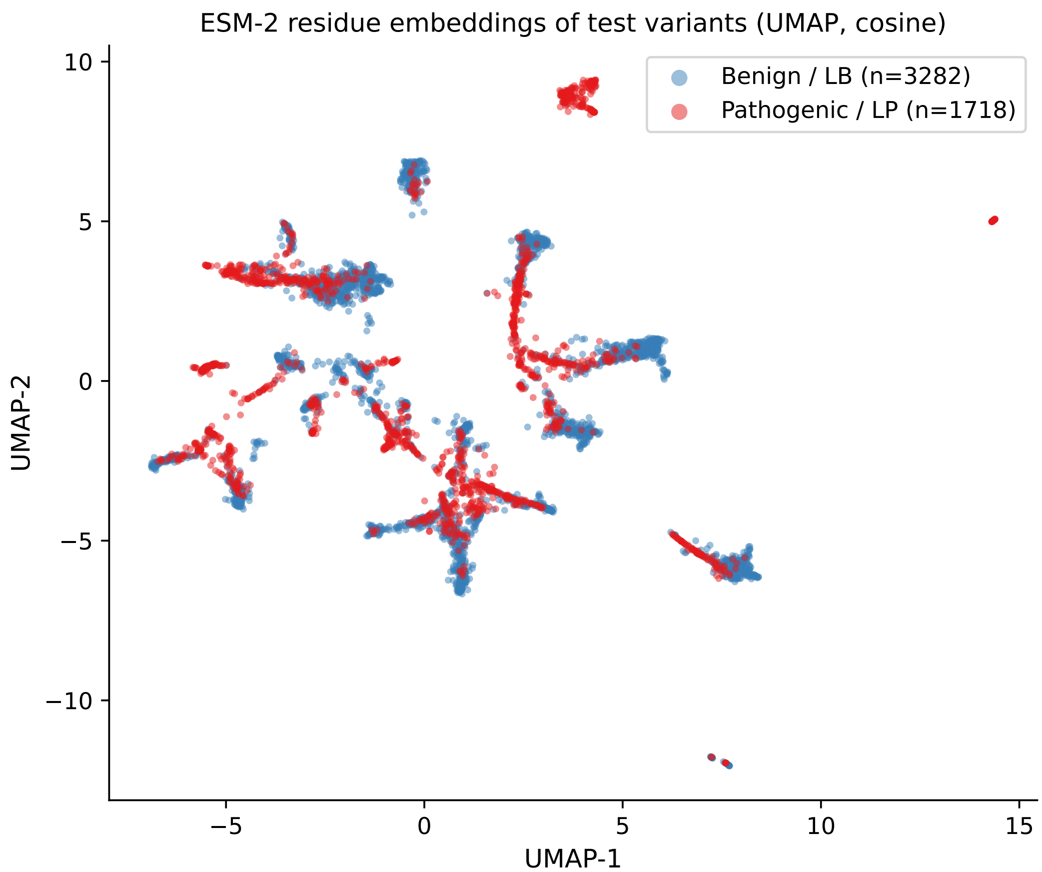

Figure A6: **UMAP of ESM-2 test embeddings.** A 5,000-variant random subsample of the test set, colored by ClinVar label. Pathogenic and benign clouds are partly separable in the embedding space even before any classifier is trained – the head's job is largely a linear cleanup.

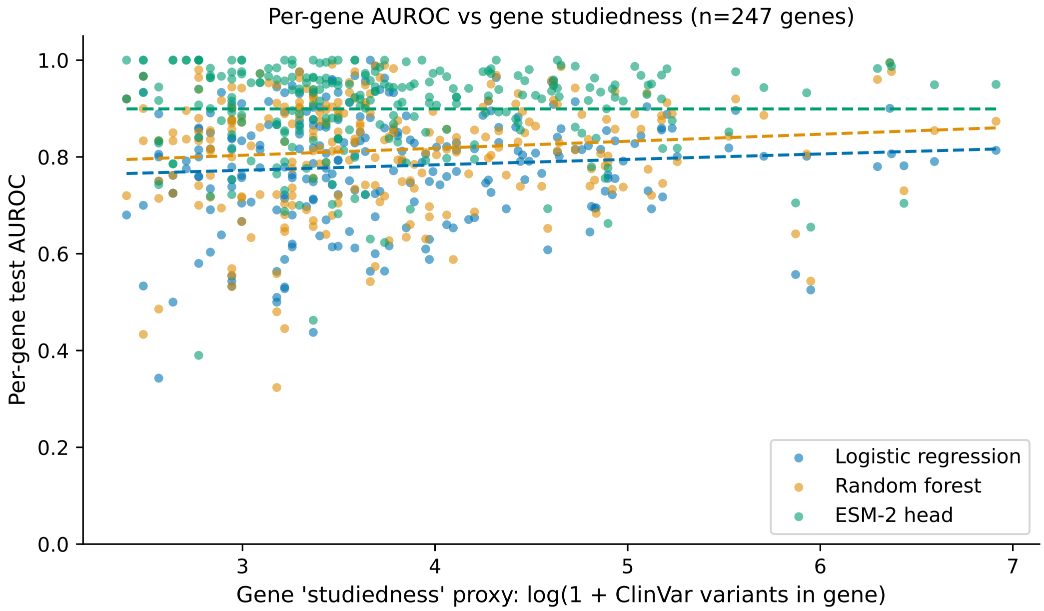

Figure A7: **Per-gene AUROC vs gene studiedness.** A linear fit of per-gene AUROC versus $\log(1+\text{ClinVar variants per gene})$ (a proxy for how heavily a gene is studied). The ESM-2 head's slope is not noticeably steeper than the baselines', suggesting its advantage is broadly distributed rather than concentrated on well-represented genes.

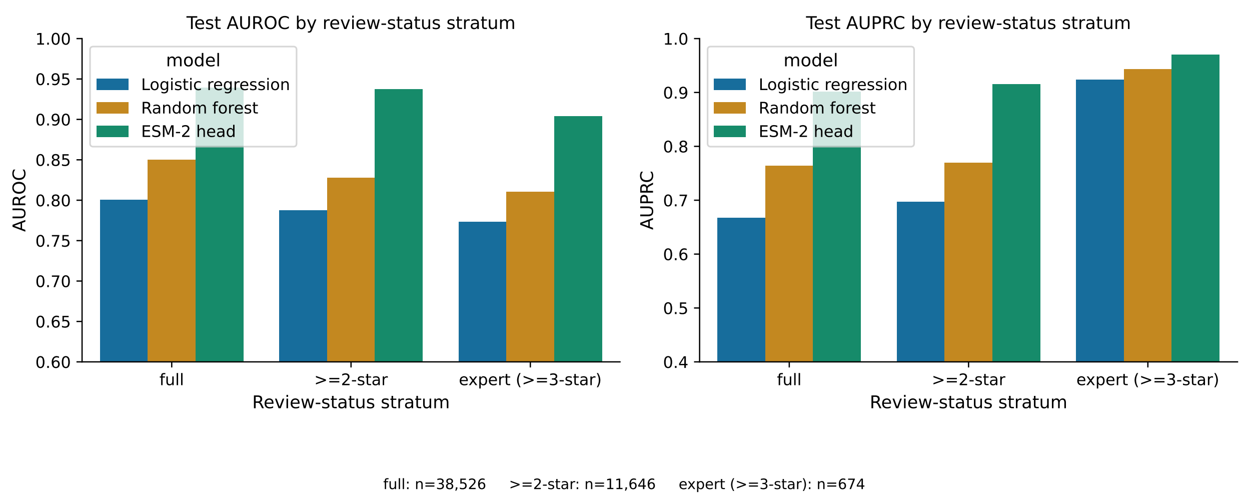

Figure A8: **Review-status sensitivity.** Test AUROC and AUPRC stratified by ClinVar review-status confidence: full retained set, $\geq$2-star (criteria, multiple submitters, no conflicts), and expert ($\geq$3-star: reviewed by expert panel or practice guideline). Conclusions hold across strata; the small expert-only stratum is more variable.